CliffordClock: a Cl(1,3) spacetime-algebra pipeline for field-gradient and DC-Stark frequency-shift budgeting in optical lattice clocks

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Optical lattice and ion clocks target systematic-uncertainty budgets at the 10^{-18} level, where DC-Stark shifts from residual stray electric fields, and the spatial variation of those fields across an extended atomic cloud, are a routinely underdocumented systematic. We present CliffordClock, an open-source pipeline that takes an imported (CAD/FEA-exported) or synthetic electric-field model with real spatial structure, propagates it through an atomic ensemble (a classical Monte Carlo cloud moving through the field, or a Hermite–Gauss lattice quadrature), and reports the mean fractional-frequency shift together with the full inhomogeneous-shift budget a spatially varying field produces: the shift's ensemble spread, the resulting dephasing time T_2^*, and the clock transition's spectral line profile. A field-gradient map alone hands a lab none of these; the pipeline computes them end to end, from field export to a reported, uncertainty-quantified budget. The physical coupling is an ordinary scalar second-order DC-Stark pivot, $\Delta\nu/\nu_0 = -(\Delta\alpha/2)|E(\mathbf{r})|^2/(h\nu_0)$, using literature differential polarizabilities; this scalar evaluation fully handles spatially varying fields on its own. Three evaluation modes share this physics: a fast analytic mode for lattice-clock motional states (exact, no time integration), a trajectory mode that propagates a Monte Carlo ensemble's actual motion through a spatially varying field, and a general Cl(1,3) spacetime-algebra rotor mode, verified to agree with the scalar modes to floating-point precision on every case this paper models and serving as the formalism this project is built on for physics the scalar modes cannot express (Sec. VI's roadmap). We validate against four closed-form self-consistency cases, four literature known-answer cases (^{87}Sr, ^{171}Yb DC-Stark and second-order-Doppler formulas, rtol = 10^{-10} or better), a direct rotor/scalar cross-check, and two zero-free-parameter *reproducibility* cases: NPL's published DC-Stark shift band, and Bothwell et al.'s published mm-scale redshift slope. Two extensions widen the physical model: a multi-surface thermal environment, and a quantum-motional systematics package for trapped ions. The motional package evaluates second-order Doppler from the atom's own motional state, applies per-mode participation and intrinsic-micromotion corrections for a two-ion crystal, and computes coherent Ramsey visibility under motional squeezing. Four further cases validate these additions: an arithmetic reproduction of a published trapped-ion secular-motion total, a per-mode closure against that same paper's own published row built from published trap-drive-frequency geometry alone, a position-resolved reproduction of a published blackbody-radiation-shield scan, and a formula check against a second lab's own published exchange-factor table. Each carries its own evidentiary class, weaker than this paper's two reproducibility cases. A blind prediction, against data the pipeline's inputs did not already combine, would be the stronger claim; the project currently reports zero blind-prediction cases, and producing one is the concrete next validation goal. A showcase scenario propagates a Monte Carlo Sr-87 cloud through a finite-difference-solved chamber field with genuine curvature (two asymmetric electrodes plus a patch-potential wall spot), reporting a mean fractional shift of -9.965×10^{-17} with a 6.514×10^{-18} ensemble spread and $T_2^* = 279.7\,\mu$s, computed identically (to 1.237×10^{-16} relative) through both the trajectory and rotor evaluation modes. Code, tests, and figure-generation scripts are public under the AGPLv3.

I. INTRODUCTION

Optical atomic clocks based on neutral-atom optical lattices and trapped ions have reached fractional-frequency systematic uncertainties at or below 10^{-18} [1–3]. At this level, a clock's DC-Stark shift budget, the frequency shift from a transition's differential static scalar polarizability $\Delta\alpha$ coupling to the square of a residual stray electric field at the atoms, is no longer a formality: it is routinely one of several systematics that must be independently bounded, and its spatial *gradient* across an extended atomic ensemble (a Ramsey-Bordé cloud, a multi-site optical lattice, or an ion crystal) can broaden the clock transition line and add to the shift's own uncertainty even after the mean field is nulled. Groups building or upgrading a clock apparatus typically characterize this budget with one of three approaches: a general-purpose finite-element solver (COMSOL, SIMION, or similar) whose output is manually carried into a hand

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or spreadsheet Stark-shift calculation; a lab's own unpublished script, rewritten by each new graduate student; or, for detailed treatments of specific atomic properties (Rydberg-state Stark maps, polarizability calculations), a dedicated tool such as ARC [4], in the broader spirit of validated, community-maintained physics simulation packages such as QuTiP [5] for open quantum systems. None of these directly close the loop from an imported field-gradient map (an FEA export, or a hand-measured stray-field profile) to a reported, uncertainty-quantified fractional-frequency shift and dephasing time for a specific trapped ensemble and interrogation protocol. That is the workflow gap this project addresses.

CliffordClock is a Python pipeline that takes an electric-field model (a closed-form synthetic field, or an imported CSV/COMSOL grid with the real spatial structure a CAD/FEA export or a hand-measured survey actually has) and an atomic ensemble specification (species, trap, motional regime, interrogation time), and closes the loop that field map alone leaves open: it reports the resulting fractional frequency shift, its statistical uncertainty, *and* the full inhomogeneous-shift budget the field's spatial variation produces across the ensemble. That budget is the shift's spread, the inhomogeneous dephasing time T_2^*, and the clock transition's spectral line profile. A lab with only a field-gradient map and a hand or spreadsheet calculation gets a single mean-shift number at best; this pipeline turns the same map into the dispersion budget that map implies, for a specific trapped ensemble and interrogation protocol, without re-deriving the perturbation theory or re-implementing a numerically stable field-squared evaluation from scratch. Sec. V demonstrates this end to end on a chamber-scale field with genuine curvature. The target regime is the 10^{-18}-level DC-Stark and gradient budgeting relevant to current optical lattice clocks (worked here for ^{87}Sr and ^{171}Yb, the two $J{=}0 \rightarrow J{=}0$ species with literature-cited differential polarizabilities in the project's species registry).

Physically, this is an ordinary second-order (quadratic) Stark shift: optical-clock states carry no permanent electric dipole moment, so the field couples via the transition's differential static scalar polarizability, and this scalar coupling fully handles a spatially varying field on its own: nothing about a field gradient or curvature requires anything beyond it. What this project adds underneath is a second, independently re-derivable formulation of the same physics in the Cl(1,3) spacetime algebra of proper time and spin connections [6, 7]: a *rotor* engine (a rotor is a rotating geometric object representing the atom's internal clock phase; advancing it along the atom's *worldline*, its path through space and time, is the general relativistic-kinematics calculation this project is built on) capable of representing boost/rotation effects exactly, with no small-angle or single-order perturbative truncation in its own exponential map. Section II describes the three evaluation modes and what each is for. The scalar modes are performance tiers of one shared physical model, and the rotor mode computes the same physics through the Cl(1,3) algebra; the two agree to floating-point precision on every case this paper computes. The rotor engine earns its place through the roadmap physics of Table I. Magnetic couplings, internal-state structure, and transport in accelerating frames all carry directional or geometric content, and none of them can be written as a simple shift value at each point in space the way the DC-Stark and blackbody terms can; representing them takes the rotor's richer mathematical object.

II. PHYSICAL MODEL

A. The scalar DC-Stark pivot

The transition frequency at a point $\mathbf{r}$ in the trap, relative to its unperturbed value ν_0, is written as a dimensionless *pivot* $P(\mathbf{r})$, the local fractional rate factor multiplying proper time, such that the local proper-time rate is $d\tau = P(\mathbf{r})\sqrt{1 - v^2/c^2}\, dt$. For the physical DC-Stark coupling used throughout this paper's validated results,

$$P(\mathbf{r}) - 1 \;=\; \frac{\Delta\nu(\mathbf{r})}{\nu_0} \;=\; -\frac{\Delta\alpha}{2}\,\frac{|E(\mathbf{r})|^2}{h\nu_0}, \tag{1}$$

with $\Delta\alpha = \alpha_{\mathrm{excited}} - \alpha_{\mathrm{ground}}$ the transition's differential static scalar polarizability and ν_0 the clock transition frequency (code cross-reference: CONVENTIONS.md E14b). Equation (1) is the same textbook second-order Stark formula used throughout the optical-clock literature [8–10], and every evaluation mode in this paper (Sec. II B) computes it *directly* from the total field $E(\mathbf{r})$ at each point. There is no baseline/perturbation split at this stage (that split, $E_0/\delta E$, belongs to the field smoother upstream, Eqs. (10)–E12 below), and no restriction to a spatially uniform field anywhere in this formula: a field with a real gradient or curvature across the ensemble is evaluated at each atom's own position, exactly, by the scalar pivot alone. Sec. III shows why this direct evaluation is numerically safe and how that safety is pinned by an adversarial regression test. The instantaneous fractional rate perturbation including kinematic (second-order Doppler) time dilation is

$$\delta\tilde{\omega}(\mathbf{r}, \mathbf{v}) \;=\; P(\mathbf{r})\sqrt{1 - v^2/c^2} - 1, \tag{2}$$

(E21) integrated over an atom's trajectory to give its accumulated fractional phase, and averaged over the ensemble (uniform weights for a classical Monte Carlo sample; for a lattice motional-state ensemble, Gauss–Hermite quadrature weights, a small set of weighted sample points that exactly average polynomial functions over the atomic wavefunction) to give the reported $\langle\Delta\nu/\nu_0\rangle$ (E22–E23). For a $J{=}0 \to J{=}0$ clock transition (the lattice species ^{87}Sr and ^{171}Yb, and the $J{=}0 \to J{=}0$ ion clocks ^{27}Al$^+$ and ^{115}In$^+$, all in the species registry) the scalar $\Delta\alpha$ is the whole DC-Stark story. An ion clock whose upper state has $J \geq 1$ additionally carries a permanent electric-quadrupole moment coupling to the field *gradient* at first order; that term is implemented and described in Sec. II F, while the tensor-polarizability correction to the second-order Stark shift itself remains roadmap physics (Sec. VI). A second, linearized coupling, $P(\mathbf{r}) - 1 = \delta E(\mathbf{r}) \cdot \boldsymbol{\mu}/(m_e c^2)$ (E14a, an explicit effective dipole moment $\boldsymbol{\mu}$), is also implemented; Eq. (1) provably reduces to it under a bias-field linearization (CONVENTIONS.md E14b "Bridge identity"). E14a is not part of this paper's physical claims about a real DC-Stark shift; it is used only for the closed-form self-consistency cases of Sec. IV (Table III, rows V1–V4), where an explicit, hand-tunable coupling constant is the useful property.

B. Three evaluation modes, one physical model

Eq. (1)'s pivot is evaluated by three user-facing evaluation modes, all computing the same physical quantity, that trade generality for speed in a controlled, verified way:

- **Fast analytic mode** (`integration.mode=fast_path`, the default for lattice-clock motional states): for a static Gauss–Hermite quadrature ensemble (Sec. II A) in a time-independent field, the phase accumulated over an interrogation time T is exactly $\Delta\Phi_q = \delta\tilde{\omega}_q \tilde{T}$ per quadrature node (E29), an algebraic evaluation with no time integration whose cost is therefore independent of T.

- **Trajectory mode** (`integration.mode=direct` for `ensemble.regime=classical`, the default there): a Monte Carlo ensemble's atoms are propagated on their actual classical trajectories through the field (velocity-Verlet dynamics in the trap, E21's pivot evaluated along the way), so a spatially varying field's real structure (gradients, curvature, any shape a CAD/FEA export carries) is sampled exactly where each atom's trajectory visits it. This is the mode Sec. V's showcase runs.

- **Rotor mode** (`integration.mode=worldline`): the full Cl(1,3) spacetime-algebra engine below, advancing each atom's Lorentz frame along its worldline (its path through space and time).

For every case this paper models, the scalar modes (fast analytic and trajectory) and the rotor mode agree to floating-point precision. This is a directly verified statement: the scalar and rotor accumulators are run on *identical* trajectories/nodes and compared head to head, both in the codebase's standing test suite (`tests/test_integrator_stark_rotor.py`) and in this paper's own showcase computation (Sec. V, Fig. 6), where the two evaluations of the same Monte Carlo ensemble agree to 1.237×10^{-16} relative in the mean shift and 2.910×10^{-11} absolute in per-atom accumulated phase. The scalar modes exist purely for speed and are not an approximation of a "more correct" rotor calculation happening underneath: for every physical coupling this project currently implements, the rotor result and the scalar result are the same number, computed two independent ways.

The rotor engine is nonetheless the general formalism this project is built on, and its distinct necessity is not hypothetical: it begins exactly where a physical effect stops being expressible as a single number ($\delta\tilde{\omega}$) per point in space, which is where every item in Table I's roadmap column, and Sec. VI's extension list (magnetic couplings, internal-state structure, transport-induced geometric phases), lives. The trapped atom's local Lorentz frame is advanced along its worldline by

$$\frac{dR(\tilde{\tau})}{d\tilde{\tau}} = -\tfrac{1}{2}\,\Omega(\mathbf{r}(\tilde{\tau}))\,R(\tilde{\tau}), \tag{3}$$

(E17, times in Compton units $\tilde{\tau} = \tau/\tau_C$, $\tau_C = \hbar/m_e c^2$; R is a *rotor*, a rotating geometric object representing the atom's internal clock phase) with the *interaction bivector* $\Omega = \omega_{\text{boost}} + I\,\omega_{\text{rot}}$ (E18; a bivector is the geometric-algebra object representing an oriented plane, here the plane of the internal clock's rotation), built from the pivot's spatial gradient (a boost/frame-tilt bivector, $\omega_{0k} = \partial_k \ln P(\mathbf{r})$, E16) and the pivot's own internal rotation rate. The rotor equation is integrated with an exponential-midpoint stepper, exactly rotor-group-preserving at any step size (E19), and the phase it extracts by composing successive $\exp(\Omega)$ rotors is compared against the primary scalar phase of Eq. (2) as a standing cross-check (E24): agreement is required at first order in the boost bivector, with a genuine second-order ($O(\omega_{\text{boost}}^2)$) divergence between the two paths (visible only for gradients far steeper than any realistic lattice-clock trap) expected by construction, the rotor result being the more complete one in that regime.

An Ω-bivector construction for Eq. (1) (`build_omega_stark`, instantiating E16/E18 for the physical DC-Stark coupling) drives `integration.mode=worldline`'s true Cl(1,3) rotor directly under `coupling.type=stark_dc`. The scalar/rotor equivalence under the physical coupling is verified directly: the rotor-vs-scalar test suite (`tests/test_` `integrator_stark_rotor.py`) checks first-order agreement at realistic field/velocity scale, the permitted $O(\omega_{\rm boost}^2)$ divergence under an exaggerated boost, a uniform-field null, a $v = 0$ static-node identity, a non-symmetric-field-gradient orientation guard (ruling out an axis-transposition error in the spin-connection contraction), and step-halving convergence of the rotor accumulator under a genuinely time-varying interaction bivector along a moving trajectory (measured $\mathcal{O}(d\tilde{\tau}^2)$, ratio ≈ 4 per halving, against an endpoint-evaluated stand-in that measures ≈ 2); the NPL reproducibility case of Sec. IV is independently re-run end to end through this rotor path, reaching the identical *MET* verdict and, to float64-relative precision, the identical predicted band as the scalar case; and Sec. V's showcase re-runs a genuinely moving Monte Carlo ensemble through both paths at once.

C. The blackbody-radiation pivot term

A second contribution enters the same scalar pivot $P(\mathbf{r})$ that Eq. (1) defines for the DC-Stark shift. Any vacuum chamber above absolute zero glows faintly in the infrared: its walls radiate thermal photons, the way a warm object radiates heat across a distance. That glow is itself an electric field, and it couples to the atom's clock states through the very same differential polarizability $\Delta\alpha(0)$ the DC-Stark term already uses. The blackbody-radiation (BBR) pivot term is

$$(P-1)_{\rm BBR} = \frac{\Delta\nu_{\rm stat}\,(T/T_0)^4 + \Delta\nu_{\rm dyn}(T)}{\nu_0}, \tag{4}$$

referenced to a room-temperature baseline $T_0 = 300$ K (E32). The static coefficient is $\Delta\nu_{\rm stat} = -(\Delta\alpha(0)/2h)\langle E^2\rangle_{T_0}$, under the full energy-density convention $\langle E^2\rangle_T = (4\sigma_{\rm SB}/\epsilon_0 c)\,T^4$ [8, 11]. The equation carries no separate leading minus sign: $\Delta\nu_{\rm stat}$ is already negative, and that alone is what makes the shift negative, in exactly the same $P-1 = \Delta\nu/\nu_0$ convention Eq. (1) established.

The DC-Stark and BBR contributions then combine the simplest way possible: they add, $P(\mathbf{r})-1 = (P-1)_{\rm stark}(\mathbf{r}) + (P-1)_{\rm BBR}$ (E33). One might worry about a cross term between the static trap field and the fluctuating thermal field, but it vanishes exactly: the thermal field is isotropic, so its vector average $\langle E_{\rm BBR}\rangle$ is zero and the cross term averages away with it. The one product term that does survive, $(P-1)_{\rm stark}\cdot(P-1)_{\rm BBR} \sim 10^{-33}$, is far below anything this paper reports. The bath itself needs no stochastic treatment either: a roughly one-second interrogation averages over $\sim 10^{14}$ independent femtosecond-scale bath cycles, so the atom experiences the deterministic expectation $\langle E^2\rangle$, the same deterministic treatment every published BBR evaluation uses.

$\Delta\nu_{\rm dyn}(T)$ is not a truncated $1/T$ power series. Lisdat, Dörscher, Nosske, and Sterr proved this Taylor expansion has zero convergence radius for Sr's dominant BBR-coupled transition [11], and the historical three-term truncation left published Sr corrections low by a documented 4.7×10^{-18}. The registry instead fits $\Delta\nu_{\rm dyn}(T)$ as a small polynomial to the exact Planck integral. For ^{87}Sr this uses the PTB group's rescaling of the Lisdat shape onto Aeppli et al.'s more precise anchor [2, 12], cross-verified to 10^{-19} for $T \le 300$ K (Sec. IV E's benchmark). Because the bath is isotropic and every registered species is $J{=}0 \to J{=}0$, the term reaches the rotor engine only through this scalar pivot, with zero spin-connection gradient for uniform T (Sec. VI).

Figure 1 shows this pivot across its full validity window, with the published dataset it inherits its shape from laid directly on top: the blue curve is Eq. (4) evaluated by the real engine from 50 to 350 K with a coefficient-uncertainty band, the red points are the Lisdat, Dörscher, Nosske, and Sterr open dataset's own tabulated total BBR shift [11] converted to the same fractional units via this project's own registry clock frequency, and the black star marks the operating point Aeppli et al. at JILA published independently (Sec. IV E). Because the registry's dynamic-term polynomial is the PTB-2025 rescaling of this same dataset's fit shape onto Aeppli et al.'s more precise 2024 anchor, the curve and the dataset points trace the same shape by construction, and the lower panel's residual is not measurement noise: it is the real, expected offset that revision introduces, growing from a few parts in 10^{20} near 50 K to 167.9×10^{-19} near 350 K, with a -62.2×10^{-19} offset at the 300 K point itself, everywhere larger than the dataset's own stated uncertainty. This figure visualizes the same arithmetic-reproduction relationship Sec. IV E already documents at Aeppli et al.'s single operating point, extended across the full window; it introduces no new validation case and does not change this paper's validation headline.

D. The multi-surface thermal environment

A shielded laboratory clock sees a chamber interior held to a fraction of a kelvin of uniformity, so Eq. (4)'s single temperature T already matches every surface the atoms see. A clock deployed outside that kind of enclosure, riding in a vehicle, sitting behind a viewport a diurnal cycle can warm, or standing near a heat source, has no such guarantee, and its visible surfaces can differ by tens of kelvin. Eq. (4)'s single T is, in general, the $\langle T^4\rangle$-matched effective

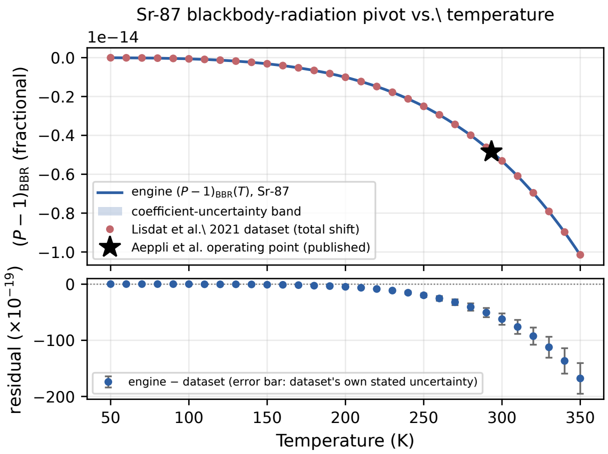

FIG. 1. The Sr-87 BBR pivot (Eq. (4)) across its 50-350 K validity window (blue curve, shaded coefficient-uncertainty band), overlaid with the Lisdat et al. 2021 open dataset's own tabulated total BBR shift (red points, converted to fractional units via this project's own registry clock frequency) and Aeppli et al.'s published operating point (black star, Sec. IV E). Bottom panel: the engine-minus-dataset residual, in units of 10^{-19}, with error bars from the dataset's own stated uncertainty. Because the registry's dynamic-term coefficients are the PTB-2025 rescaling of this dataset's own fit shape onto a more precise later anchor (Sec. II C), the residual's growing structure is the expected, real signature of that anchor revision. Regenerated by `paper/figures/fig5_bbr_temperature.py` against the real `cliffordclock.integrator.omega.bbr_pivot_perturbation` and the committed dataset fixture `benchmarks/data/lisdat2021_dataset/`.

temperature of a real, non-uniform environment, exact for the static term by construction; the dynamic term's higher powers of T carry no matching guarantee, so a T chosen to reproduce the static shift can still miss the dynamic one. The mismatch between the exact multi-surface shift and the best single-temperature approximation crosses 10^{-18}, close to a modern lattice clock's total systematic budget, at roughly an 11 K spread across two comparably sized surfaces and at roughly 19 K for a small warm viewport against a larger chamber body.

E37 replaces the single temperature with an explicit description of N surfaces, each carrying an effective solid-angle fraction $w_i = \Omega_i/4\pi$, a temperature T_i, an optional temperature uncertainty σ_{T_i}, and an optional interior emissivity ϵ_i. The fractions w_i are a lab-supplied input, from geometry, an FEA model, or a ray-traced exchange-factor calculation, and they are enforced to sum to one at parse time and again at evaluation time. Writing T_0 for E32's 300 K reference temperature and $M_n = \sum_i w_i\,(T_i/T_0)^n$ for the environment's n-th weighted moment, E32's static and dynamic terms are evaluated against these moments in place of a single power of T/T_0:

$$(P-1)_{\mathrm{BBR}} = \frac{\Delta\nu_{\mathrm{stat}}\,M_4 + \sum_n c_n\,M_n}{\nu_0}, \tag{5}$$

with c_n the same per-species dynamic-term coefficients Eq. (4) already uses (E37). M_4 matches the environment's true static shift exactly by construction, a weighted power mean of the surface temperatures; the higher moments M_6, M_8, M_{10} generally do not equal $M_4^{n/4}$ for a non-uniform environment, and that divergence is what a T chosen only to match the static shift misses. Each moment defines its own per-moment effective temperature $T_{\mathrm{eff},n} = T_0\,M_n^{1/n}$, and every pipeline report prints all of them, so a user reads the divergence from the report text without recomputing it from the shift alone. A uniform environment (one surface, $w_1 = 1$) makes every M_n equal to $(T_1/T_0)^n$ exactly, so Eq. (5) reduces to Eq. (4) term for term, and the two paths agree bit for bit by construction.

A reflective enclosure changes how a lab's own raw solid-angle fractions convert to effective weights. Nosske et al. model their transportable strontium clock's atoms as sitting inside one reflective enclosure of interior emissivity ϵ, pierced by one or more apertures leaking in a different temperature; the enclosure's own reflections give that leaked-in radiation more chances to reach the atoms than its raw geometric solid angle alone suggests, so their published closed form for one aperture of raw fraction w is $w_{\mathrm{eff}} = w/(w + (1-w)\epsilon)$ [12], reducing to $w_{\mathrm{eff}} = w$ at $\epsilon = 1$ and growing past w as the interior becomes more reflective. E37 carries this topology: at most one surface in an environment may set an emissivity, that surface is the enclosure, and every other surface is a direct-view aperture. Writing W for the apertures' combined raw fraction, each aperture's effective fraction is $w_{i,\mathrm{eff}} = w_i/(W + (1-W)\epsilon)$, the published formula applied to the combined W and split across the individual apertures in proportion to their own share of it. The enclosure's own effective fraction is the complement $1 - \sum_i w_{i,\mathrm{eff}}$ of the apertures' fractions, fixed by the same two-temperature derivation that fixes the aperture fractions themselves; for a single aperture this reduces to the published formula exactly, and Sec. IV J checks that reduction against the paper's own numbers.

Per-surface temperature uncertainties σ_{T_i} propagate through the same analytic-derivative pattern Eq. (4)'s own uncertainty note uses, scaled by each surface's effective weight, in two combination modes. The independent mode, the default, treats the surfaces' temperature errors as uncorrelated and combines them in quadrature. The correlated mode treats them as moving together, a single shared calibration-chain error affecting every sensor coherently, the motivation Aeppli's 2025 JILA thesis gives for combining its own several correlated temperature estimates by linear pooling; every registry coefficient here carries the same sign, so the correlated band is never narrower than the independent one. Every surface's own temperature must lie in the 50–350 K window Eq. (4) already validates against, enforced individually so a composite result draws only on temperatures the underlying fit was checked against. A surfaces list is given inline in the pipeline configuration or, equivalently, as a path to a plain-text surfaces table via the `--radiation-surfaces` CLI flag, and the two forms parse to the identical structure.

E. The gravitational-redshift pivot term and extended samples

General relativity contributes the next pivot term, and it needs no electromagnetic field at all: two clocks at different heights in Earth's gravity tick at different rates. For a site or atom at height $h(\mathbf{r}) = \hat{u}\cdot\mathbf{r}$ along a configured up direction $\hat{u}$, relative to a configured reference height h_{ref}, the weak-field clock-rate factor enters the pivot as

$$(P-1)_{\mathrm{grav}}(\mathbf{r}) \;=\; \frac{g\left(h(\mathbf{r}) - h_{\mathrm{ref}}\right)}{c^2}, \tag{6}$$

the leading term of the metric proper-time ratio (E36); the next metric order is smaller by another factor of $g\,\Delta h/c^2$ and is irrelevant at any laboratory scale. The sign is pinned to the physical statement that a higher clock runs faster: $(P-1)_{\mathrm{grav}} > 0$ for $h > h_{\mathrm{ref}}$, in the same $P - 1 = \Delta\nu/\nu_0$ convention every other pivot term uses, and a shipped regression test asserts exactly this statement. The magnitude at standard gravity is $g/c^2 = 1.0911370 \times 10^{-16}$ per metre of height.

Composition follows the same additive pattern as the BBR term (E33): $P(\mathbf{r}) - 1 = (P-1)_{\mathrm{stark}}(\mathbf{r}) + (P-1)_{\mathrm{BBR}} + (P-1)_Q(\mathbf{r}) + (P-1)_{\mathrm{grav}}(\mathbf{r})$, with the quadrupole term of Sec. II F present for ion species. The gravitational term depends only on position in the potential, so its genuine cross term with any field-driven shift is of order 10^{-31}, far below anything this paper resolves. One consistency point deserves its own sentence: Eq. (6) and the kinematic time-dilation factor of Eq. (2) are the two leading terms of one proper-time expansion, the potential term and the velocity term, and the pipeline adds each exactly once, with no $g \times v$ cross term at any order it resolves.

The term earns its keep in the `lattice_extended` ensemble regime, which distributes n_{sites} copies of the single-site lattice quadrature (the exact Hermite–Gauss motional average of Sec. II A) along a configured axis at a configured spacing, weighted by a Gaussian or uniform site-occupation envelope. Every extended-lattice node is a static point exactly like a single-site lattice node, so the fast analytic mode stays exact and the rotor cross-check applies unchanged, and each site's own position feeds every pivot term in scope: the local field through the Stark term, the uniform BBR term, and Eq. (6)'s height dependence. The output is a per-site frequency map, the observable Bothwell et al. published [13]: each site's position, occupation weight, and weighted mean fractional shift, together with a weighted least-squares linear fit across the site means whose slope is the map's headline number.

An extended sample forces one labeling discipline the single-site regimes never needed. The frequency spread across sites is dominated by the deterministic linear gradient: higher sites tick systematically faster, a repeatable and in principle refocusable structure, and folding it silently into a T_2^* or linewidth would misread deterministic inhomogeneity as stochastic decoherence. Every extended-lattice report therefore carries a test-pinned note stating that its ensemble spread and T_2^* include the deterministic gradient, and the site map reports both the total spread and the gradient-removed residual spread (the weighted standard deviation of the site means after subtracting the best-fit line), extending the showcase's SEM-versus-T_2^* discipline (Sec. V) to the deterministic-versus-stochastic axis.

Treating g as uniform has a stated validity range. The error of a single-g evaluation grows quadratically with the sample's height extent, through the height dependence of g itself, and reaches the 10^{-19} fractional floor only near a 76 m extent; the pipeline warns at 10 m, an order of magnitude under that bound, and documents that beyond laboratory scale the physically correct input is a surveyed geopotential difference. Eq. (6) reports height differences within the sample relative to a local reference, and the whole apparatus's absolute geoid and Earth-rotation offset is a separate constant outside its scope. At the 10^{-19} level the choice of g itself also stops being a formality: standard gravity is a defined placeholder, and the lab's own surveyed local value is the physically correct input. For Boulder, Colorado, the surveyed $g = 9.796$ m/s^2 [14] gives a slope of 1.0900×10^{-19} per millimetre against standard gravity's 1.0911×10^{-19}, a difference well inside Bothwell et al.'s own stated prediction uncertainty (Sec. IV D).

F. Ion-clock electric-quadrupole shift

Ion clocks bring a genuinely new coupling into the same pivot. A D- or F-state upper clock level with $J \geq 1$ carries a permanent electric-quadrupole moment Θ, and that moment couples to the electric-field *gradient* at first order, in contrast to the DC-Stark and BBR terms, which are second order in the field itself. The level shift is

$$\Delta E_Q(J, m_J) \;=\; \frac{\Theta_{\rm SI}}{2}\,\frac{J(J+1) - 3m_J^2}{J(2J-1)}\;\hat{n}^{\mathsf{T}} G(\mathbf{r})\,\hat{n}, \tag{7}$$

(E34) with $\hat{n}$ the quantization-axis unit vector and $G(\mathbf{r})$ the traceless symmetric part of the field-gradient tensor $\partial_i E_j$ the field smoother already delivers. The m_J factor and the overall sign follow Roos et al.'s Eq. (1), transcribed directly from primary text [15], and the coordinate-free $\hat{n}^{\mathsf{T}} G\,\hat{n}$ contraction is an exact rewriting of the published axial-plus-asymmetry angular form, derived in the project's conventions record and cross-checked numerically against that angular form as a standing test.

For a fixed (J, m_J) and quantization axis the whole angular factor is a scalar coefficient, so the shift composes additively into the pivot alongside every other term (E35), with no cross term: the quadrupole coupling is first order in the gradient while the Stark and BBR terms are second order in the field, different field quantities at different orders. One exact identity comes free with the tensor structure. For any three mutually orthogonal quantization axes the three contractions sum to the trace of G, which is zero by construction, so averaging a measurement over three perpendicular field orientations cancels the quadrupole shift exactly, for any gradient tensor; the pipeline implements this as a supported averaging mode and tests the null at machine precision. The engine's null is the ideal one: a real clock's residual after such averaging comes from imperfect axis orthogonality and timing, a separate systematic the engine does not model, and no report presents the exact cancellation as an achievable experimental bound.

The species registry carries the measured quadrupole moments this term needs: the $D_{5/2}$ states of Ca^+, Sr^+, and Ba^+ and the $D_{3/2}$ and $F_{7/2}$ states of Yb^+, each with its literature citation and verification status recorded as registry fields. The Yb^+ $F_{7/2}$ entry's negative Θ doubles as the registry's sign anchor: a shipped regression test requires it to produce an opposite-sign shift from any positive-Θ D state under an otherwise identical gradient, m_J, and axis. The $J{=}0 \to J{=}0$ ion clocks Al^+ and In^+ carry no first-order quadrupole moment at all and run through the scalar DC-Stark path of Sec. II A, with a report note stating that their small hyperfine-mediated second-order quadrupole residual is out of scope.

Every ion-species report also carries a binding boundary statement, because a quadrupole map alone would overstate what has been characterized. The same stray DC field that sources the DC-Stark and quadrupole terms also displaces an ion from its trap's RF null and drives excess micromotion, a separate shift pathway the engine does not model, and published measurements in a real Sr^+ trap [16] attribute roughly 95% of the m_J-dependent budget to micromotion-driven tensor Stark shifts against roughly 5% from the quadrupole term alone. The report note states this shared-cause structure explicitly, so a quadrupole map is never presented as a stray-field budget (Sec. VI places the tensor-polarizability and micromotion package on the roadmap together for exactly this reason).

G. Quantum-motional second-order Doppler

Eq. (2)'s kinematic term carries the second-order Doppler shift along a classical trajectory's own real, nonzero velocity, but a lattice or trapped-ion atom evaluated at a static quadrature node (Sec. II B, $v = 0$ exactly) has no classical velocity for that term to act on. The atom still moves: its own zero-point and thermal motional-state population gives it a nonzero velocity *expectation*, and the second-order-Doppler shift from that motion is frequently the largest single systematic in a published trapped-ion clock budget. For an atom or ion held near its motional ground state in N normal modes indexed i, each with an ordinary (not angular) mode frequency f_i, as reported by resolved-sideband thermometry, and a mean vibrational occupation number $\bar{n}_i$, the mode angular frequency is $\omega_i = 2\pi f_i$ and the velocity-variance expectation over the motional state is

$$\langle v^2 \rangle = \sum_i \frac{\hbar\omega_i}{m}\, p_i \Big(\bar{n}_i + \tfrac{1}{2}\Big), \tag{8}$$

with m the species mass resolved from the same registry Eq. (1) uses and $p_i \in (0, 1]$ a per-mode participation factor (default 1, discussed below). The fractional shift is Eq. (2)'s same second-order-Doppler form, here evaluated as an expectation value over the quantum motional state in place of a classical instantaneous velocity, $(P-1)_{\rm motional} = -\langle v^2 \rangle/(2c^2)$ (E38). The $+\frac{1}{2}$ zero-point term keeps this shift nonzero even at $\bar{n}_i = 0$ for every mode: an atom cooled exactly to its motional ground state still carries the irreducible quantum-mechanical motional-Doppler floor Eq. (8) predicts.

This term composes into the pivot without double-counting the classical kinematic term Eq. (2) already carries. A classical Monte Carlo ensemble's trajectories already sample a real, nonzero velocity at every point, and $\sqrt{1-v^2/c^2}-1$ along that trajectory is already the complete second-order-Doppler shift for that ensemble's own motion; composing Eq. (8) on top of it there would add the same physics twice, and the pipeline rejects a motional-state configuration paired with a classical ensemble at config-parse time for exactly this reason. The lattice and extended-lattice regimes instead evaluate at static quadrature nodes with $v = 0$ exactly, where the classical kinematic term is zero identically, so Eq. (8) supplies new physics at those nodes and nothing already counted there gets counted again. The motional state is spatially uniform across the ensemble, one state per run, matching Eq. (4)'s own single-temperature convention: its spatial gradient is zero, it reaches the pivot's denominator only, and it composes additively alongside every other pivot term with no cross term at this order.

A two-ion crystal divides a shared normal mode's motion between its two ions by the squared components of the mass-weighted eigenvectors, so the clock ion does not carry a mode's entire $\langle v^2\rangle$ contribution when it shares that mode with a different-mass partner; p_i above is exactly that squared eigenvector component. For a two-ion crystal's axial pair, the standard normal-mode solution gives p_i in closed form as a function of the mass ratio $\mu = m_{\text{partner}}/m_{\text{clock}}$ alone (Wübbena et al. [17]), an in-phase (center-of-mass, COM) and an out-of-phase (stretch, STR) mode whose equal-mass limit gives $p = 1/2$ for both, each ion carrying exactly half. The two radial pairs additionally depend on trap RF/DC geometry parameters this closed form does not take as input; a caller who has the trap's own measured radial mode frequencies instead reconstructs the clock ion's radial participation by inverting the coupled two-ion radial eigenproblem against those frequencies. The Coulomb curvature that eigenproblem needs comes from the axial confinement alone: the two ions share one axial spring constant k_z (the same DC field gradient sets a mass-independent per-ion spring constant), and expanding the inter-ion Coulomb repulsion to second order about the equilibrium spacing gives a transverse curvature exactly half the axial one in magnitude and opposite in sign, so $c = k_z/2$ with no permittivity constant surviving (Wübbena et al. [17], Eq. 7). The resulting quadratic eigenproblem's branch ambiguity is resolved by the rule that the lighter ion carries the higher bare radial frequency, since the RF pseudopotential that confines both ions scales as $1/m$ at fixed trap drive.

A radial pair's own intrinsic micromotion, the RF-driven oscillation that accompanies ordinary secular motion at any nonzero trap RF drive, is distinct from excess micromotion (the paragraph below) and multiplies the participation-weighted term above through a per-axis leading-order enhancement factor

$$F_{\text{axis}} = 1 + \frac{q^2}{2a_{\text{axis}} + q^2}$$

(Berkeland et al. [18], Eq. 10), equal to 2 exactly when the trap's Mathieu a parameter vanishes for that axis and departing from 2 as the trap's DC asymmetry splits a_x from a_y. The clock ion's own leading-order Mathieu parameters q, a_x, a_y, a_z solve from the trap's published RF drive frequency together with the axial Coulomb curvature and the two reconstructed clock-ion bare radial frequencies, via $\omega_i = (\Omega/2)\sqrt{a_i + q_i^2/2}$ (Berkeland et al. [18], Eq. 9, with $q_z = 0$ along the axial direction) and the Laplace constraint $a_x + a_y = -a_z$ that holds for any static quadrupole DC potential: two equations in two unknowns, zero degrees of freedom, with no separate trap-geometry parameter supplied by hand. Because a_i and q_i are each linear in $1/m$ at fixed drive voltage, geometry, and charge, the clock ion's solved parameters mass-scale to predict the partner ion's own bare radial frequencies, an independent, falsifiable check whose result Sec. IV I reports.

A lab's own measured characterization of excess micromotion, the ion's forced oscillation at the trap's RF drive frequency when a stray DC field displaces it from the RF null, enters as a second, independent input channel: already reduced by the lab to an equivalent rms velocity $v_{\text{rms,emm}}$, it adds in quadrature to Eq. (8)'s $\langle v^2\rangle$ sum. The RF trap dynamics that produce excess micromotion, the stray-field-induced displacement from the RF null and the resulting time-varying velocity at the drive frequency, remain a roadmap package (Sec. VI); this channel consumes a lab's own characterization exactly as Sec. II D consumes a lab's own solid-angle fractions.

H. Coherent Ramsey visibility

Each worldline's accumulated perturbation phase $\Delta\Phi_k$ (Sec. V) composes with the rotor formalism of Sec. II B: identifying the engine's internal-circulation plane bivector $\hat{B}_C$ with the imaginary unit, $\exp(\Delta\Phi_k\,\hat{B}_C)$ builds a unit rotor R_k carrying that worldline's phase, the rotor-algebra restatement of the ordinary complex phase factor $e^{i\Delta\Phi_k}$ every standard treatment of Ramsey fringes already uses. This construction builds R_k from the full accumulated phase $\Delta\Phi_k$, the same phase Sec. V already accumulates and reports. The composed dynamical rotor Eq. (3)'s stepper produces along the way carries a different angle: its own rotor factor advances at half the physical rate, the same spinor convention that lets a double-sided sandwich $R\,v\,\tilde{R}$ recover the full rotation for an ordinary vector, so reading a phase off that rotor's own angle would understate the true Ramsey phase by a factor of two.

An ensemble's coherence is the population-weighted coherent sum of these per-worldline rotors,

$$M = \sum_k p_k \, R_k,$$

with p_k the ensemble's existing probability weights (uniform for a classical Monte Carlo sample, quadrature weights for a lattice motional-state ensemble); writing c and s for M's scalar and $\hat{B}_C$-plane components, the ensemble's fringe visibility and phase are $V = \sqrt{c^2 + s^2}$ and phase $= \arg(c + is)$. M is deliberately never renormalized back to a unit rotor: summing the group elements linearly and taking the modulus afterward is what lets $V < 1$ represent genuine decoherence. Two alternative constructions erase that signal outright. Averaging the per-worldline phases before exponentiating the mean, $|\exp(i \sum_k p_k \Delta\Phi_k)|$, gives modulus 1 for any phase spread, because the spread information is discarded before the modulus is taken. Renormalizing M to a unit rotor forces modulus 1 by construction regardless of the ensemble's true spread; both constructions are encoded as kill tests that must disagree with the real combiner on a spread ensemble.

For Gaussian-distributed accumulated phases, $\Delta\Phi_k \sim \mathcal{N}(\mu, \sigma_\Phi^2)$, the characteristic-function identity $\mathbb{E}[e^{i\Delta\Phi}] = e^{i\mu - \sigma_\Phi^2/2}$, the same identity behind this project's own inhomogeneous-dephasing-time definition (Sec. V), gives the closed form

$$V = \exp\left(-\sigma_\Phi^2/2\right) \tag{9}$$

exactly, the validation identity a live pipeline run confirms (`tests/test_coherent_visibility.py`; a squeezing sweep in `notebooks/13_trapped_ion_quantum_motion.ipynb`, Sec. 5). Thermal, coherent, and squeezed motional states are the Gaussian-distributed accumulated-phase states a classical-ensemble sampler can draw real worldlines from. Fock states with $n \geq 1$ and cat states sit outside that class, since a classical sampler cannot draw a positive-probability worldline ensemble from either; this worldline-ensemble representation stops at Gaussian motional states, and no report from this module quotes a visibility for a non-Gaussian one. The classical ensemble sampler accepts an optional per-axis squeezing parameter r (`ensemble.squeezing_r`), scaling the trap-frame position-quadrature variance by e^{-2r} and the velocity-quadrature variance by e^{+2r}, the engine's own sign convention for which quadrature squeezing narrows; the same sampled draw feeds both the pre-existing mean-shift pipeline and the visibility calculation here, so the two never see independently resampled ensembles.

I. Scope: what is and is not modeled

TABLE I. Modeled vs. not-modeled physics (current codebase scope). Every "not modeled" row is a documented, intentional MVP scope boundary, not an oversight; see Sec. VI and benchmarks/MAPPING.md's engine-scope recap.

Modeled	Not modeled
DC-Stark shift, scalar $\Delta\alpha$ (Eq. (1))	AC lattice-light (magic-wavelength) shift
Second-order (kinematic) Doppler (Eq. (2))	Zeeman shifts (1st or 2nd order)
Blackbody radiation, uniform T (Eq. (4), 50–350 K)	Per-atom BBR solid-angle map across an extended cloud
Multi-surface BBR thermal environment (Eq. (5))	
Gravitational redshift across an extended sample (Eq. (6))	BBR M1/E2 multipoles ($\sim 6 \times 10^{-20}$), hyperpolarizability
Ion electric-quadrupole shift, D/F-state ions (Eq. (7))	Tensor polarizability and excess-micromotion dynamics
Quantum-motional 2nd-order Doppler (Eq. (8))	
Coherent Ramsey visibility, Gaussian motional states (Eq. (9))	Non-Gaussian visibility (Fock $n \geq 1$, cat states)
Field-gradient inhomogeneous dephasing (T_2^*)	Atom-atom collisional / density shifts
Classical (Monte Carlo) motional ensembles	Lattice band-structure / tunneling
Lattice and extended-lattice (Hermite–Gauss quadrature) ensembles	Probe (clock-laser) AC Stark shift
^{87}Sr, ^{171}Yb; Ca$^+$/Sr$^+$/Ba$^+$/Yb$^+$ D/F states; Al$^+$/In$^+$ (scalar)	Time-dependent (RF trap / probe) fields
Imported (CSV/FEA) or synthetic fields	

III. NUMERICAL METHODS

A. Precision discipline

The target regime is fractional shifts at 10^{-17}–10^{-19}, frequently on top of stray-field biases many orders of magnitude larger. The pipeline guards structurally against two catastrophic-cancellation failure modes. First, the absolute Compton phase ($\omega_C T \sim 10^{20}$ rad over a one-second interrogation) is never accumulated; every integration path accumulates only the *perturbation* quantity $\delta\tilde{\omega}$ (dimensionless, of order the fractional shift itself), with compensated (Kahan) summation in every long accumulation loop. Second, the DC-Stark field-squared term $|E|^2$ of Eq. (1) is evaluated *directly* from the total field $E(\mathbf{r})$ (`cliffordclock.integrator.omega.pivot_perturbation_stark`, the rate function every scalar evaluation mode shares, with no baseline/perturbation split supplied), not via a term-by-term expansion. This is numerically safe for a structural reason: unlike the absolute-Compton-phase case above, Eq. (1) contains no subtraction of large near-equal quantities in the first place. The shift is a single product chain, $|E|^2$ multiplied once by the small prefactor $\Delta\alpha/(2h\nu_0)$, so there is no cancellation for a term-by-term decomposition to protect against. A separate, term-by-term decomposition of the same formula, $|E|^2 = |E_0|^2 + 2E_0 \cdot \delta E + |\delta E|^2$ (E11's field split), *is* implemented in the library as `stark_pivot_terms`, for callers that need the gradient-driven cross term $2E_0 \cdot \delta E$ isolated from the baseline as its own quantity (e.g. a gradient-only systematic-budget line item); this paper's reported shifts do not use it.

This safety claim is pinned by an adversarial pipeline-level regression test defined in `tests/ test_e2e.py` as `test_wp9_ major1_ stark_dc_ adversarial_ gradient_ pins_no_ cancellation_ regression` (recapped in `docs/validation. md`). The test constructs a $|E_0| = 10^5$ V/m uniform bias along with a field gradient tuned so the shift differential between the lattice quadrature's extreme nodes is $\sim 10^{-19}$, then compares each node's directly-evaluated shift against a 50-digit `decimal` reference: the measured per-node absolute error is $\sim 1.3 \times 10^{-26}$, seven orders of magnitude below the 10^{-19}-level signal it resolves. An embedded discrimination guard confirms the test is not vacuous: applying the same comparison to the genuinely dangerous antipattern (forming each node's $O(1)$ pivot $P = 1 + \text{(shift)}$ explicitly and recovering the differential by subtracting two such totals) degrades to a $\sim 5 \times 10^{-20}$ error, which the test asserts (and confirms) must exceed its passing bound. This also distinguishes two numbers a reader could otherwise conflate. The 10^{-18} fractional-shift scale motivating this precision discipline (Sec. I) is the physical *target regime*. The bounds this paper measures are (i) relative agreement with closed-form and literature references at rtol $= 10^{-10}$ or better (Table III) and (ii) the absolute-error pin above, $\sim 10^{-19}$–10^{-26}, in this specific adversarial configuration.

These bounds span nineteen orders of magnitude and mix relative ratios with absolute fractional-shift errors, which is exactly the condition under which a single number gets misquoted as "the accuracy of the tool." They are therefore never combined onto one scale in this paper, and each carries its character wherever it is quoted: the closed-form and literature rows of Table III are relative agreements (rtol $\leq 10^{-10}$), the adversarial per-node error pin is an absolute fractional-shift error (`tests/test_e2e.py`'s MAJOR-1 regression pin, with the same test also recording the far larger error the subtract-two-near-unity-pivots antipattern would produce), the rotor/scalar agreement of Sec. II B and Sec. V is exactly zero relative in the showcase's ensemble mean shift, and the 10^{-18} physical regime is a design target the pipeline is built for, never something this paper measured. No one of these numbers substitutes for any other: a reader who wants "the accuracy of a reported shift" should read the validation table and this paper's stated target regime together, not either alone.

B. Field model

An imported or synthetic field is decomposed as

$$E_{\text{total}}(\mathbf{r}) = E_0(\mathbf{r}) + \delta E(\mathbf{r}), \tag{10}$$

with E_0 a least-squares degree-1 (uniform + linear) analytical baseline and δE the residual, fit by a thin-plate-spline radial basis function per vector component (E12); the field gradient $\partial_i E_j$ (E13) is obtained by exact automatic differentiation of the fitted evaluator, so the field and its gradient are always mutually consistent by construction. The RBF fit is performed with SciPy; the gradient-consistent evaluator is re-implemented in JAX [19] so it composes with the pipeline's own autodiff-based downstream code (Sec. II B's spin connection, and the second-order Doppler kinematic term).

C. Performance tiers within the fast analytic and trajectory modes

The scalar rate integrand $\delta\tilde{\omega}(\mathbf{r}(t), \mathbf{v})$ contains no Compton-frequency content (the fast $\sim 10^{20}$ rad/s carrier is removed analytically by the perturbation discipline above), so its time variation is set entirely by the atom's own motional timescale, microseconds to milliseconds for realistic traps. This is what makes real (microsecond-to-second) interrogation times tractable at all, and Table II summarizes the resulting cost structure of the three evaluation modes (Sec. II B) plus one internal performance variant within the trajectory mode. The fast analytic mode is exact for a static motional state in a time-independent field: every quadrature node has $\mathbf{v} = 0$ and a constant rate, so the time integral is trivial and the cost is independent of the requested interrogation time T. The trajectory mode's direct (default) variant steps the classical integrator at a resolution set by the trap period, with

$$d\tilde{\tau} \;=\; \frac{T_{\mathrm{orb}}}{N_{\mathrm{res}}\,\tau_C}, \qquad N_{\mathrm{res}} = 100 \text{ (default)}, \tag{11}$$

(E31) with $T_{\mathrm{orb}} = 2\pi/\min(\omega_{xyz})$ the slowest trap axis's period. A second trajectory-mode variant, *secular averaging*, replaces per-step time integration with a single average over one orbital period (one full cycle of the trap's own periodic motion) scaled by the full interrogation time; it applies whenever the classical motion is periodic in an isotropic trap. The rotor mode's Compton-scale integrator remains available as the first-principles cross-check every other mode/tier is validated against.

TABLE II. The three evaluation modes and their performance tiers. Cost is stated in the requested interrogation time T.

Evaluation mode	Tier	Regime	`integration.mode`	Basis	Cost in T
Fast analytic	A	lattice (motional state)	fast_path (default)	exact (E29)	$O(1)$
Trajectory	B(i)	classical	direct (default)	scalar integrator, E31 step	$O(\text{steps})$
Trajectory (periodic)	B(ii)	classical, isotropic trap	secular	one-orbit average (E30)	$O(1)$
Rotor	C	either	worldline / Compton-fine direct	Compton-scale Cl(1,3) integrator	$O(T/\tau_C)$

An accuracy study (Fig. 2, reproducing the CONVENTIONS.md V4 closed-form harmonic-trap case: a linear-gradient field's oscillating pivot contribution cancels exactly over full trap periods, leaving the trap-center pivot value plus the closed-form second-order-Doppler time average) sweeps the step size at fixed N_{res} and measures a convergence order of 1.999 against the exponential-midpoint stepper's design order of 2. At the default $N_{\mathrm{res}} = 100$, the phase matches the closed form to $2.84e-09$ relative, with rotor-norm drift $1.14e-13$, both comfortably inside this project's rtol $\leq 10^{-8}$ / drift $< 10^{-12}$ target bounds.

D. Verification methodology

Beyond the closed-form and literature comparisons of Sec. IV, two structural verification mechanisms protect against a codebase that merely agrees with itself. First, the Cl(1,3) geometric product and bivector exponential kernel are cross-checked at the algebra level against the third-party open-source `clifford` package [20] (the pygae project's general geometric-algebra library on PyPI), used purely as an independent test-oracle reference in CI, disjoint from this project's own structure-tensor implementation. Second, a plain-NumPy reference implementation (`tests/reference_impl.py`) re-derives the phase-accumulation formulas directly from the equations recorded in CONVENTIONS.md, sharing no code path, constants module, or numerical kernel with the pipeline itself (`cliffordclock.integrator`, `cliffordclock.cl13`); the pipeline's scalar and rotor phases are each checked against it independently (Table III rows V3, E24).

IV. VALIDATION

A. Labeling rules (binding)

This section follows the labeling discipline established for this project's benchmark work (`benchmarks/RESULTS.md`, `benchmarks/MAPPING.md`): a case is reported as **reproducibility** when every pipeline input traces to an independent publication and the resulting predicted value reconstructs a number the cited source(s) already combined those same inputs to compute; it is reported as **blind prediction** only when the pipeline predicts a measured quantity from inputs that were not already combined, by anyone, to produce that measured value. These are categorically distinct

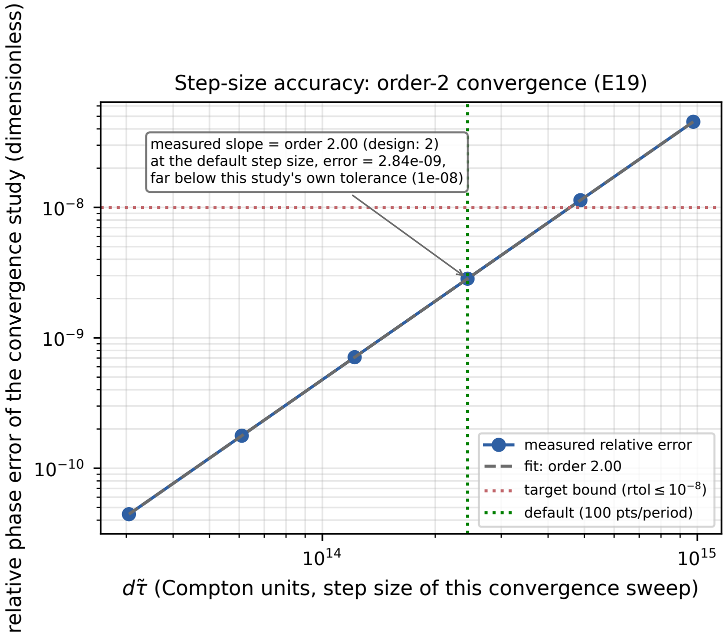

FIG. 2. This figure is a numerical convergence study of the Cl(1,3) rotor stepper (Sec. II B's exponential-midpoint integrator, E19): relative phase error against the V4 closed form, swept over the stepper's step size $d\tilde{\tau}$ (auto-selected via Eq. (11)). The measured log-log slope (1.999) matches the stepper's design order of 2, and the annotation marks that at the default step size the error is $2.84e - 09$, comfortably below this study's own rtol $\leq 10^{-8}$ tolerance. Regenerated by `paper/figures/fig3_step_size_accuracy.py` directly against `cliffordclock.integrator.worldline`, using the same trap/field/coupling scenario as `notebooks/02_step_size_study.ipynb` and `tests/test_fastpath_select_dtau.py`. This is a measurement of the integrator's own convergence order, not the pipeline's real-world shift accuracy; the bounds that actually characterize a reported shift's precision are those of Table III and the precision discipline of Sec. III A.

claims, never conflated in this paper: a reproducibility *MET* verdict demonstrates end-to-end pipeline correctness against independently published inputs and outputs with no fitted or tuned parameter anywhere in the chain (a real, falsifiable claim: a sign error, a unit-conversion bug, or a wrong $\Delta\alpha$-to-k_S derivation would show up as a predicted band that does not overlap the published one) but it does *not* demonstrate that this engine can predict a clock shift nobody had already computed.

Two weaker labels complete the taxonomy. An **arithmetic reproduction** runs a published row that is itself computed, never independently measured, back through the engine's own formula and coefficients (Sec. IV E); a **cross-vintage comparison** predicts a published measured quantity from an input of an independent vintage and method, one the published result never combined, while some other ingredient of the comparison still traces to the source's own apparatus calibration (Sec. IV F). The cross-vintage class sits strictly between its neighbors: weaker than a blind prediction, because not every ingredient on both sides is independent of every other, and stronger than an arithmetic reproduction, because the compared-against number was never derived from the predicting input. Neither weaker class ever joins the reproducibility or blind-prediction counts.

As of this writing the project has two reproducibility cases (Secs. IV C and IV D) and zero blind-prediction cases; no case in this paper, this codebase's test suite, or its benchmark reports is described as "validated against an independent measurement" where that framing would imply the stronger, currently unmet claim.

B. Validation table

Table III is generated directly by `paper/figures/table_validation.py` against the real pipeline, species registry, and `benchmarks/` code. Every "Computed" entry is a live pipeline run, and every "Reference" entry is either a closed form evaluated independently in this script, an independent-NumPy reference module, a literature formula applied to the project's own cited species data, or (for the NPL row) the published band of Sec. IV C.

TABLE III. Validation summary: case, reference value/method, this pipeline's computed value, and measured agreement. V1–V4 are closed-form self-consistency cases (this engine against its own analytically solvable configurations and, for V3, an independent plain-NumPy re-implementation); KA1–KA4 are literature known-answer cases (textbook DC-Stark and second-order-Doppler formulas, using this project's cited $\Delta\alpha$ values); E24 is the rotor/scalar cross-check of Sec. II B; NPL and Bothwell are the two experimental reproducibility cases (Secs. IV C and IV D); Roos is the cross-vintage quadrupole-slope comparison of Sec. IV F, whose *NOT MET* verdict is the expected outcome (it recovers the literature's own Θ theory-versus-measurement tension) and which joins neither headline count. Generated by `paper/figures/table_validation.py`; see that script and `tests/test_e2e.py` / `tests/test_known_answers.py` for the exact configurations reproduced.

Case	Reference	Computed	Agreement
V1: uniform field, rotor vs. scalar	closed form (CONVENTIONS.md §9): 0	0	PASS
V2: constant gradient, static atom	3.77×10^{-10}	3.77×10^{-10}	rel. err. 0 (PASS, rtol 10^{-12})
V3: quadrupole field, M=100 (scalar)	independent NumPy reference	1.07×10^{-13}	rel. err. 2.35×10^{-16} (PASS, rtol 10^{-10})
V4: harmonic trap, linear gradient	closed form (SHM + 2nd-order Doppler)	order 1.999 (design: 2)	rel. err. 2.84×10^{-9} at N_{res}=100
E24: rotor/scalar cross-check	independent NumPy reference	1.07×10^{-13}	PASS
KA1: Sr87 uniform-field DC Stark	textbook formula, Middelmann et al. 2012: -7.17×10^{-17}	-7.17×10^{-17}	PASS (exact to fp64)
KA2: Yb171 uniform-field DC Stark	textbook formula, Sherman et al. 2012: -3.50×10^{-17}	-3.50×10^{-17}	PASS (exact to fp64)
KA3: Yb171 gradient shift + spread	Gaussian-moment reference: mean -4.90×10^{-18}, $\sqrt{\mathrm{Var}}$ 4.66×10^{-24}	mean -4.90×10^{-18}, $\sqrt{\mathrm{Var}}$ 4.66×10^{-24}	rel. err. mean 0, var 5.45×10^{-11} (rtol 10^{-8})
KA4: Sr87 second-order Doppler	equipartition $-3k_BT/2mc^2$: -7.98×10^{-21}	$-8.00 \times 10^{-21} \pm 6.33 \times 10^{-23}$ (SEM)	0.32σ (PASS, bound 5σ)
NPL: Sr87 reproducibility case	Bowden et al. 2017: $[-3.20 \times 10^{-20}, -1.20 \times 10^{-20}]$	$[-3.29 \times 10^{-20}, -1.21 \times 10^{-20}]$	bands overlap (*MET*; reproducibility, not blind prediction)
Bothwell: Sr87 mm-scale redshift	Bothwell et al. 2022: -9.8×10^{-20}, -1.28×10^{-19} per mm	-1.0900×10^{-19} per mm	0.48σ / 0.70σ (*MET*; reproducibility, not blind prediction)
Roos: Ca^+ two-ion quadrupole slope	Roos et al. 2006: 2.975 $\mathrm{Hz\,mm^2/V}$	3.1152 $\mathrm{Hz\,mm^2/V}$ (Itano-2006 theory Θ)	+4.71% (*NOT MET*, expected: known Θ tension; cross-vintage comparison)

C. The NPL reproducibility case

Bowden et al. (NPL) [10] independently measured a residual stray electric field at their ^{87}Sr lattice-clock atoms via Rydberg-state electrometric interferometry (a completely different atomic transition and spectroscopic method from the clock transition itself) and converted it to a clock-transition DC-Stark shift using the Middelmann et al. differential polarizability [8], the same literature value this project's ^{87}Sr species-registry entry cites. **No parameter was fitted at any stage: the field magnitude and its uncertainties (NPL's Rydberg-spectroscopy measurement), the differential polarizability (PTB's measurement), and the clock frequency are each independently published constants.**

NPL could have computed this number themselves, and in a sense already did: for a spatially uniform 1.52 V/m field the shift is a single evaluation of Eq. (1), so this case is not evidence of predictive power beyond the published record. What it demonstrates instead is that the entire pipeline (field ingestion, lattice quadrature, precision-safe numerics, asymmetric uncertainty propagation, and reporting) reproduces an independently published result end-to-end with no fitted or tuned parameter anywhere in the chain. Following this project's benchmark taxonomy (`benchmarks/RESULTS.md`, `benchmarks/MAPPING.md`; see the labeling rules above), that makes this a **reproducibility** case: NPL already combined the same two published ingredients this pipeline combines, so reconstructing their number checks end-to-end pipeline correctness against independently published inputs and outputs. The stronger claim, predictive power on a measurement nobody had already converted to a shift, belongs to the blind-prediction category, and it remains unmet.

NPL's residual field, $E = 1.52$ V/m, carries independent, individually asymmetric statistical and systematic uncertainties. These are combined in quadrature separately on each side (never symmetrized against each other, and never treated as Gaussian), giving field bounds $[1.298, 2.142]$ V/m. The pipeline (`coupling.type: stark_dc`, the lattice fast path, a genuine 1 s interrogation; the same machinery Table III's KA1 row validates) is run three separate times at this field's low, nominal, and high bounds, giving the predicted band $[-3.290 \times 10^{-20}, -1.208 \times 10^{-20}]$, compared in Fig. 3 against NPL's own published band $[-3.200 \times 10^{-20}, -1.200 \times 10^{-20}]$. The two bands overlap (a precisely defined closed-interval overlap test); NPL's nominal value falls inside the predicted band and vice versa. **Verdict: *MET* (reproducibility case, with no fitted parameters anywhere in the chain).**

NPL's own single-point field measurement does not exercise this pipeline's actual differentiator: gradient-driven shifts, inhomogeneous broadening, and line shapes computed from a field with real spatial structure. Eq. (1) at a uniform field is a one-line calculation regardless of what pipeline performs it. Sec. V demonstrates that capability directly, end to end, on a chamber-scale field with genuine curvature; what it does *not* yet demonstrate is that capability against a field characterization nobody has already converted to a shift. Producing and validating that class of result against an independently supplied field is exactly the blind-prediction program described in Sec. VI, and the showcase's scenario remains, like the NPL case above and the closed-form validation suite, a demonstration whose every input is a documented scenario choice.

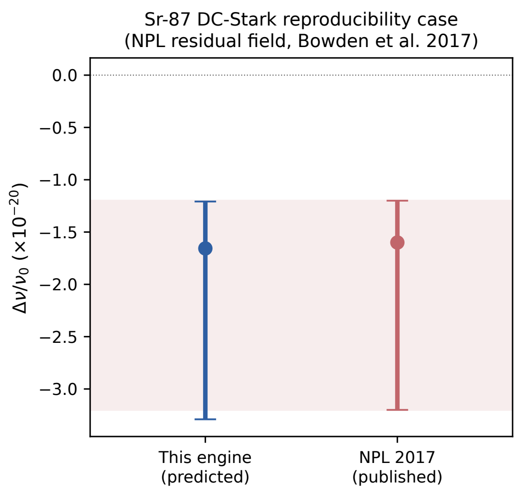

FIG. 3. The NPL reproducibility case: this pipeline's predicted DC-Stark shift band (from NPL's independently measured residual field and this project's Middelmann-sourced $\Delta\alpha$) vs. NPL's own published band [10]. Shaded region: NPL's published band. Error bars: asymmetric field/shift uncertainty, propagated per-side in quadrature (no Gaussian assumption on the asymmetric field uncertainty). Regenerated by `paper/figures/fig2_npl_band.py` via `benchmarks/run_benchmarks.run_npl_reproducibility_case`.

D. The Bothwell millimetre-scale redshift reproducibility case

The second reproducibility case targets a different pivot term and a different kind of observable. Bothwell et al. resolved the gravitational redshift across a single millimetre-scale ^{87}Sr lattice sample, the first single-apparatus measurement of gravitational time dilation at this length scale, publishing a per-pixel frequency map and two independently analyzed, systematics-corrected linear slopes [13]. The extended-lattice pipeline of Sec. II E is configured to their published sample geometry: a Gaussian site-occupation envelope with $\sigma = 402.7\,\mu$m inferred from their stated $\pm 1.5\sigma$ analysis window, 5945 computational sites at the real 406.5 nm magic-lattice spacing, and the surveyed local gravity for Boulder, $g = 9.796$ m/s^2 (van Westrum's NOAA survey [14], the reference Bothwell et al. themselves cite), the physically correct input at this level (Sec. II E).

Run end to end, the per-site machinery produces the map of Fig. 4 and a weighted-least-squares slope of -1.0900×10^{-19} per millimetre in Bothwell et al.'s own coordinate convention. Their axis puts lower physical positions at larger coordinate, so their published gradients are negative while the engine's own height convention gives a positive slope;

the benchmark states and applies this sign mapping explicitly, so the sign agreement is deliberate. Against the two published corrected slopes, method A's $-9.8 \times 10^{-20} \pm 2.3 \times 10^{-20}$ per millimetre (a fourteen-dataset campaign) and method B's $-1.28 \times 10^{-19} \pm 2.7 \times 10^{-20}$ per millimetre (a synchronous two-region analysis), the prediction lands within 0.48σ and 0.70σ respectively, and the band verdicts are *MET* against method A and *MET* against method B.

The comparison isolates exactly the term it should. The published corrected slopes already have per-pixel density and second-order-Zeeman corrections applied before the linear fit, plus budget-level corrections for BBR, lattice light, DC Stark, and pixel calibration, so the corrected number targets the isolated gravitational gradient Eq. (6) predicts. The benchmark's own field is configured to exactly zero for the same reason: no Stark, BBR, or quadrupole term is active, so the case isolates the gravitational term to match its target. Bothwell et al.'s own DC-Stark gradient row is in-scope physics for this engine, and it enters the benchmark report as a narrative cross-reference only, since the comparison target already has it corrected out.

The binding classification is **reproducibility**, with a caveat inverted from the NPL case's. The g/c^2 arithmetic is textbook and Bothwell et al. computed it themselves trivially, so nothing about the coefficient demonstrates predictive power; what this case validates end to end is the extended-sample machinery, the per-site geometry, the envelope weighting, and the map assembly, producing the right measured-map slope with zero adjustable inputs. NPL's caveat ran the other way: there the differential-polarizability reconstruction carried the domain expertise, and the pipeline's contribution was the end-to-end evaluation. Together the two cases make the project headline of Sec. IV, two reproducibility cases.

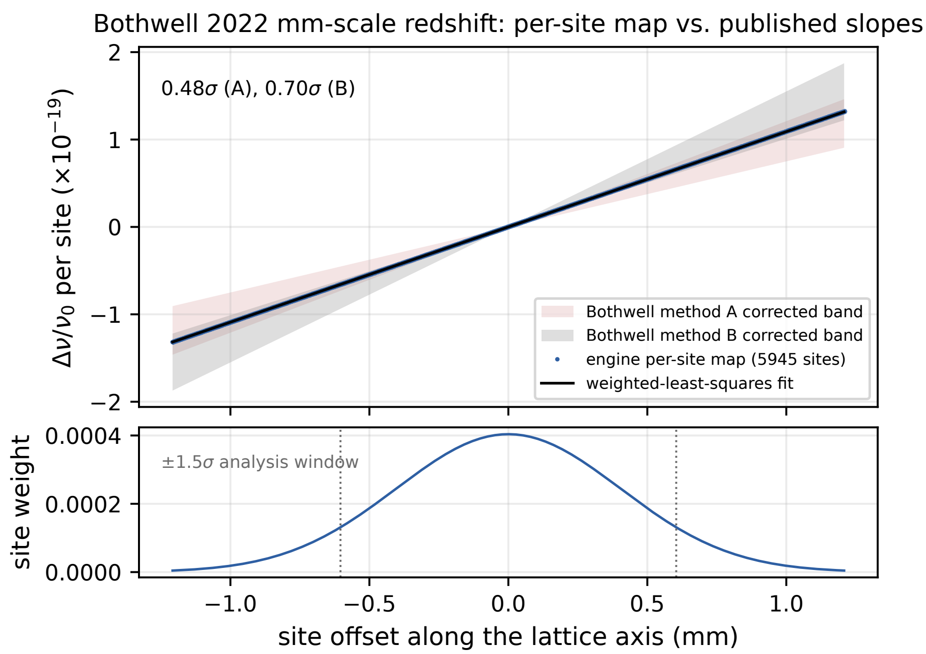

FIG. 4. The Bothwell reproducibility case: the engine's per-site fractional-frequency map (5945 sites, blue points), its weighted-least-squares slope fit (black line), and Bothwell et al.'s two published corrected slope bands [13] drawn through the same intercept (shaded wedges), all in the engine's physical-height sign convention. The annotation restates the two sigma distances (0.48σ against method A, 0.70σ against method B). Bottom panel: the Gaussian site-occupation envelope, with the $\pm 1.5\sigma$ analysis window marked. Regenerated by `paper/figures/fig7_bothwell_sitemap.py` against the real `lattice_extended` pipeline at the exact configuration `benchmarks/run_bothwell_redshift.py` pins.

E. A weaker case: the Aeppli et al. (JILA) BBR arithmetic-reproduction check

The BBR term (Sec. II C) carries a deliberately weaker check than Sec. IV C's NPL case: whether the engine arithmetically reproduces a BBR shift a source group already computed from its own measured temperature and coefficients. Aeppli et al. report their ^{87}Sr clock's in-vacuum resistance-thermometer (RTD) temperature, $T = 293.282 \pm 0.004$ K, and separately publish the BBR-row shift it produces [2]. Running Eq. (4) at that temperature with the registry coefficients gives $(P-1)_{\mathrm{BBR}} = -4.84174 \times 10^{-15}$, with a combined coefficient-plus-temperature band of $[-4.84248\times10^{-15}, -4.84101\times10^{-15}]$. Aeppli et al.'s published value spans $[-4.84245\times10^{-15}, -4.84099\times10^{-15}]$, leaving a residual of -2.251×10^{-20}; the bands overlap (verdict *MET*).

This agreement is expected almost by construction: the registry's Sr-87 dynamic coefficients are anchored to Aeppli et al.'s own value through the same PTB rescaling Sec. II C describes, so the case reproduces a paper its own inputs trace to. Nothing about Aeppli et al.'s shift was independently measured the way NPL's field was. Their published row is itself an arithmetic evaluation from their own inputs. Following this project's binding labeling discipline (`benchmarks/RESULTS.md`, the project's theory sign-off record (G7), B5), that makes this an **arithmetic reproduction of a published standard-formula evaluation**: weaker than this paper's reproducibility cases, and weaker still than the (unmet) blind-prediction category. It does *not* join the validation headline, now two reproducibility cases, NPL and Bothwell (Secs. IV C and IV D). What the case does establish is that the engine's BBR arithmetic and coefficient-provenance chain are correct: any 10^{-19}-class number here is **arithmetic-reproduction fidelity, not BBR accuracy**.

F. A cross-vintage comparison: the Roos two-ion quadrupole slope

The quadrupole term of Sec. II F carries its own benchmark case, against the measurement that anchors the field. Roos et al. prepared an entangled two-ion Ca^+ state, measured its parity-oscillation frequency as a function of an applied, mechanically calibrated axial field gradient, and fit a line with slope $a = 2.975(2)$ Hz mm^2/V [15]. The engine predicts that slope from Itano's independent theory value $\Theta = 1.917\, ea_0^2$ [21] through its own factor chain: the m_J factors of Eq. (7) evaluated by real engine calls, the doubling of the gradient each ion sees from the presence of the other, and the resulting two-ion enhancement over a single ion's shift, whose engine-derived ratio of 4.8 structurally reproduces Roos et al.'s stated 24/5.

The predicted slope is 3.1152 Hz mm^2/V, a residual of +4.71% against the measured value, and the band test returns *NOT MET*. That verdict is the expected outcome, and the case record says so in a self-contextualizing note it carries in its own JSON: the residual reproduces the literature's own tension between Itano's computed $\Theta = 1.917\, ea_0^2$ and Roos et al.'s measured $\Theta = 1.83(1)\, ea_0^2$, so the comparison validates the engine's factor chain while isolating the coefficient the two vintages disagree on.

A second, deliberately circular variant closes the loop on the arithmetic. Feeding the engine Roos et al.'s own extracted $\Theta = 1.83(1)\, ea_0^2$, the very value their paper derives by inverting the same Fig. 4a fit through $\Theta = (5/12)\, h\, a$, predicts 2.9738 Hz mm^2/V, a −0.04% residual whose bands overlap (*MET*). This variant is labeled an arithmetic reproduction, circular by construction, and its value is the factor-consistency check it performs: the engine's independently derived coefficient chain reproduces Roos et al.'s own published conversion relation exactly, confirmed by a closed-loop round trip at float64 precision.

The headline variant carries the taxonomy's cross-vintage-comparison label (Sec. IV's labeling rules), and the label keeps its two weaknesses visible: Roos et al.'s applied gradient is itself calibrated through their own trap model, and their own fit produced the very slope being predicted. It is nonetheless a genuine external comparison, distinct from the circular variant above, because Itano's Θ was never an ingredient of the fit it predicts. Neither variant joins the paper's headline counts, and the project remains at two reproducibility cases and zero blind predictions.

G. The covered slice of one platform's published budget

The validation cases above were built one systematic at a time; this subsection reads them the way a lab would, as rows of one platform's evaluation. The JILA Sr system publishes both of the evaluations this paper reproduces from published inputs, so Table IV collects the covered rows of that single platform side by side with the lab's own numbers, each row in one unit convention, each published value carrying its stated uncertainty, and each comparison reduced to a difference and a distance in units of that uncertainty. Two readings carry the table. The BBR difference sits at 0.03σ of the published uncertainty (0.02σ against the published and prediction bands combined in quadrature), small by construction since the registry's dynamic coefficients are anchored to this group's own measurement, which is exactly why that row keeps the weaker arithmetic-reproduction label. The two redshift slopes are Bothwell et al.'s own independent analysis methods, and their central values differ from each other by $+3.0\times10^{-20}$/mm (0.85σ combined), more than either differs from this pipeline's single prediction, which lies between them. The one in-scope row without a comparison, their DC-Stark gradient, is exactly the shape of the blind-prediction program of Sec. VI: comparing against it requires a residual-field characterization the paper does not publish, and a lab sharing that characterization turns this context row into the record's first blind prediction. The composed, executable form of this table is `notebooks/11_real_budget_slice.ipynb`, which also prints the report's own exclusion notes naming the rows this release does not model.

TABLE IV. The covered slice of one platform's published evaluation (the JILA Sr system: Aeppli et al. [2], Bothwell et al. [13]), computed from published inputs by the same case objects the benchmark suite commits. σ is the difference in units of the published value's stated uncertainty, a conservative convention that ignores this tool's own prediction band; for the BBR row, the only one where that band is comparable to the published one, the combined-uncertainty metric reduces the distance to 0.02σ. The context row is in-scope physics (Eq. (1)) awaiting the lab-side field characterization described in Sec. VI. Regenerated by `paper/figures/table_budget_slice.py`; the executable form is `notebooks/11_real_budget_slice.ipynb`.

Row	Published	This tool	Δ (tool−pub.)	σ	Verdict / class
BBR, fractional (Aeppli et al. Table I)	$-4.841720 \times 10^{-15} \pm 7.3 \times 10^{-19}$	$-4.841743 \times 10^{-15}$	-2.3×10^{-20}	0.03	*MET*; arithmetic reproduction
redshift slope, method A (10^{-20}/mm)	-9.8 ± 2.3	-10.90	-1.10	0.48	*MET*; reproducibility
redshift slope, method B (10^{-20}/mm)	-12.8 ± 2.7	-10.90	$+1.90$	0.70	*MET*; reproducibility
DC-Stark gradient (Bothwell et al. Table 1)	$+0.3(0.2)\times10^{-20}$/mm	awaiting a field characterization	n/a	n/a	context row

H. The Al-ion secular-motion arithmetic reproduction

Trapped-ion clocks carry a second-order-Doppler systematic from the ion's own residual secular motion that is frequently the largest single line in a published budget, and Eq. (8) is this project's first quantitative check of that term against a real trapped-ion evaluation. Marshall et al. report a $^{27}\mathrm{Al}^+/^{25}\mathrm{Mg}^+$ two-ion crystal's six normal-mode frequencies and their sideband-thermometry-measured occupations [22], and separately publish the secular-motion budget row those same inputs combine to. Running Eq. (8) at those six modes, with the Al^+ registry mass carried through every mode ($p_i = 1$ throughout, the single-species-mass form), gives $(P-1)_{\mathrm{motional}} = -115.09(2.71) \times 10^{-19}$, the propagated uncertainty coming from the published per-mode occupation uncertainties alone (Marshall et al. publish no per-mode frequency uncertainty), against their own published $-114.6(3.8) \times 10^{-19}$. The two bands overlap at 0.10σ, verdict *MET*.

This agreement is a total-level check: Marshall et al.'s six modes are two-ion normal modes, and the physically complete per-mode evaluation would divide each mode's motion between the two ions by their own normal-mode amplitudes, a quantity the single-species-mass formula above does not consume. Summed over the complete six-mode set, the naive per-mode terms still reproduce the published total inside both uncertainty bands, even though the individual per-mode terms differ from Marshall et al.'s own published per-mode row by up to several-fold. Following this project's binding labeling discipline (Sec. IV's labeling rules), that makes this an **arithmetic reproduction of a published standard-formula evaluation**: it validates Eq. (8)'s formula and unit chain end to end against a published six-mode input set and a published total. This case carries its own class, and the project's headline count stays at two reproducibility cases and zero blind predictions (Sec. IV). Sec. IV I checks the same six modes one at a time against Marshall et al.'s own published per-mode row and closes most of the gap this total-level check leaves open.

I. Per-mode participation and the intrinsic-micromotion enhancement

A close total-level agreement can hide compensating per-mode errors as easily as it can confirm the underlying formula mode by mode, so this project checks Sec. IV H's same six modes one at a time against Marshall et al.'s own published per-mode row before treating the total as settled. Applying the two-ion participation factors of Sec. II G moves the two axial modes to within a few tenths of a percent of Marshall et al.'s own published per-mode values, confirming the closed form's own derivation, but the four radial modes land at roughly half their published magnitude, ratios 0.42, 0.35, 0.52, and 0.50 (predicted over published, X-COM/X-STR/Y-COM/Y-STR). Reconstructing the radial participation from the measured X and Y mode-frequency pairs, inverting the coupled two-ion radial eigenproblem against the Coulomb curvature Sec. II G recovers from the axial spring constant, reaches essentially the same near-half ratios: each mode pair's two participations sum to one exactly regardless of how the pair's total splits between its COM and STR members, so a correctly reconstructed split moves the per-mode ratios without moving either mode pair's own total by much.

The identified reason for the near-half shortfall is a physics omission. Brewer et al.'s predecessor evaluation of the same $^{27}Al^{+}/^{25}Mg^{+}$ species pair states in a table footnote that its own published transverse per-mode values already include the shift from intrinsic micromotion, the RF-driven oscillation that accompanies ordinary secular motion at any nonzero trap drive [23], and Marshall et al.'s radial rows carry the same physics. A correction that supplies only the clock ion's participation share therefore predicts close to half of the published radial shift, because Berkeland's leading-order enhancement factor $F_{\text{axis}} = 1 + q^2/(2a_{\text{axis}} + q^2)$ (Sec. II G) equals two exactly when the trap's Mathieu a parameter vanishes along that axis and stays close to two whenever a stays small, the regime both of this crystal's radial branches occupy.

Solving for the clock ion's own Mathieu parameters from the trap's published 70.86 MHz RF drive frequency, the axial Coulomb curvature, and the two reconstructed clock-ion bare radial frequencies gives $q = 0.1913$, $a_x = -0.00588$, $a_y = 0.00230$, sitting inside Berkeland's own stated leading-order regime. Before using these parameters for anything, they meet an independent, falsifiable test: mass-scaling them to the partner ion predicts its own bare radial frequencies, a quantity nothing in the clock-ion solve's own inputs ever touches, at -0.57% (X) and -0.53% (Y) against the separately reconstructed values, both comfortably inside the few-percent band the published mode frequencies' own three-significant-figure reporting precision supports. Multiplying the reconstructed radial participations by the resulting per-axis enhancement factors, $F_x = 2.474$ and $F_y = 1.888$, moves the four radial per-mode ratios to 1.04, 0.86, 0.98, and 0.94, an improvement on every individual mode, and moves the corrected total to $(P-1)_{\text{motional}} = -1.064 \times 10^{-17}$ with a propagated uncertainty of $\pm 3.373 \times 10^{-19}$, landing at 1.62σ from Marshall et al.'s published band, bands that nearly touch without quite overlapping ($NOT\ MET$). Figure 5 shows all four totals side by side against the published band. The residual that remains is consistent with the next-order Mathieu correction Berkeland's leading-order treatment omits, of order q^2 and a few percent, amplified on the X axis by its own a_x sitting smaller in magnitude than a_y, which makes the enhancement factor more sensitive to a there.

The same construction closes an independent second dataset. Applying it to Brewer et al.'s own published trap, a different RF drive frequency (40.72 MHz) and different mode frequencies on the same species pair [23], moves the four radial per-mode ratios against Brewer et al.'s own published row (already including the transverse intrinsic-micromotion shift, per that paper's own table footnote) into a range between 0.88 and 0.93, the same qualitative closure a second, independent trap confirms; the over-determination check passes there too, at -0.95% (X) and -0.79% (Y). Brewer et al.'s Table S2 publishes a confidence-interval bound on the zero-point occupation combined with a per-mode heating rate through a time-dependent model, an input shape Eq. (8)'s static-occupation formula cannot consume. Reproducing Brewer et al.'s own total-level secular-motion row would need that time-dependent model re-implemented, the same missing-input reason that leads Sec. IV H to build its own total-level case from Marshall et al.'s inputs.

Sec. IV H's single-mass total already landed inside Marshall et al.'s published band, at 0.10σ, without either correction this section develops. That agreement is a near-cancellation of two omissions moving in opposite directions: a plain participation split alone under-predicts the radial modes by close to a factor of two, an intrinsic-micromotion enhancement factor near two applied on its own over-predicts by close to the same factor, and the single-mass formula implicitly sets both factors to one, a product of exactly one that happens to sit close to the true combined value. Figure 5's four bars trace this cancellation: the single-mass total sits inside the published band by construction, the participation-only totals (the mass-ratio closed form and the spectrum-reconstructed inversion alike) undershoot it by the same near-factor-of-two the per-mode ratios above show, and the participation-times-enhancement total recovers most of the distance the participation-only step gives up. This progression carries its own arithmetic-reproduction class throughout (Sec. IV H); none of its four totals joins this paper's reproducibility or blind-prediction headline count.

J. The PTB position-resolved thermal-environment validation

Eq. (5)'s enclosure-and-apertures topology reduces to Nosske et al.'s own published two-surface formula exactly, character for character, for a single aperture (Sec. II D). The strongest test of that reduction is the paper's own position-resolved scan, which measures a differential shift as the atoms move from the shield's center out through one of its two end apertures and compares it against their own closed-form prediction at every scanned position, going beyond a check at one fixed operating point. Nosske et al. publish enough apparatus geometry to replay that scan from the paper alone: both aperture radii to six micrometers, the shield's 20 mm length, the interior coating emissivity $\epsilon_{\text{in}} = 0.926(43)$, and the shield and room temperatures during the scan [12]. Feeding the paper's own solid-angle formula for the atom's axial position between the two apertures through Eq. (5)'s enclosure-and-apertures path at the shield center, and the pure single-temperature reduction of Eq. (4) for the far-outside limit where the aperture's effective weight saturates at one, gives a predicted differential shift of -3.325×10^{-15}.

Nosske et al.'s own closed-form model predicts $-3.32(7)\times10^{-15}$ for the same differential, and their measurement gives $-3.33(3)\times10^{-15}$; the engine's prediction, built from nothing but the paper's own published geometry and

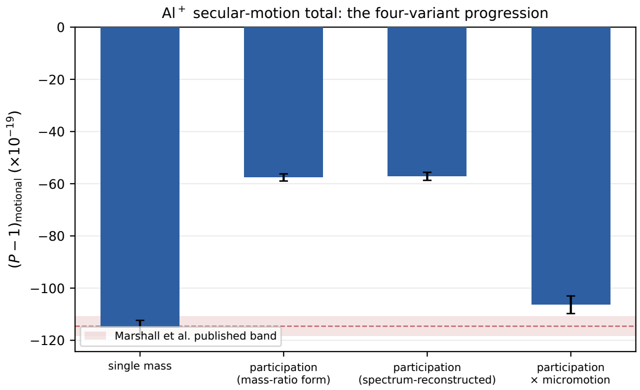

FIG. 5. The Al$^+$ secular-motion progression (Secs. IV H and IV I): four successive corrections to the same six-mode total, computed by the real engine against Marshall et al.'s published mode frequencies and occupations [22], plotted against Marshall et al.'s own published secular-motion band (shaded). The first variant carries one species mass through every mode. The second applies the two-ion participation factor from the Al$^+$/Mg^{25+} mass ratio alone (Wübbena et al. [17]). The third keeps that axial factor and reconstructs the radial pairs from the measured X/Y mode spectrum. The fourth multiplies the third's participation by Berkeland's mode-specific intrinsic-micromotion enhancement [18]. Error bars are each case's own propagated 1-sigma uncertainty. Regenerated by `paper/figures/fig8_motional_progression.py` against the real `benchmarks/run_motional_al_ion.py` case objects.

temperatures, lands within 5.011×10^{-18} of their model and 4.989×10^{-18} of their measurement, comfortably inside either one's own quoted uncertainty. This is a position-resolved formula validation: it checks the same $\Omega(z)/4\pi$ solid-angle formula, the enclosure-and-apertures reflection correction, and the per-surface engine evaluation against a real measured shift curve as the atoms move through many positions, going beyond the single-operating-point formula-transcription check the two-surface reduction alone provides. A complete shift reproduction from independently published inputs would additionally need a full shield-temperature reconstruction this project's own inputs do not attempt, so this case carries its own class in this project's benchmark taxonomy, weaker than the reproducibility class Secs. IV C and IV D carry: it joins neither headline count.

A second, independent check runs the same E37 machinery on a real lab's own published exchange-factor table. Bothwell's PhD thesis derives the exchange-factor formalism $T_i^4 = \sum_j F_{i\to j} T_j^4$ from first principles and ray-traces it across a real strontium-clock chamber, publishing the resulting factor for seven named surfaces [24]. Feeding the six controlled surfaces through Eq. (5) at the thesis's own 22°C setpoint gives $(P-1)_{\mathrm{BBR}} = -4.971 \times 10^{-15}$, agreeing with the thesis's own bookkeeping at that same setpoint, -4.972×10^{-15}, to 1.336×10^{-18}; every included surface sitting at the thesis's own single assumed temperature collapses the six-surface environment to that one temperature exactly, so this agreement checks the engine's arithmetic against the thesis's own numbers more than it tests genuine spatial non-uniformity. The seventh surface, an oven nozzle at 848 K, sits far outside Eq. (4)'s validated 50–350 K window, and Eq. (5)'s per-surface check correctly rejects it; its own static-term bound, -6.819×10^{-18}, sets the precision either number in this comparison can claim, larger than the agreement itself. Notebook 12 carries the full chamber diagram, the position-resolved extension of this same exchange-factor table, and every input this paragraph's numbers draw from.

V. SHOWCASE: A CHAMBER-SCALE FIELD'S INHOMOGENEOUS SHIFT BUDGET

This section is the pitch the rest of the paper builds toward: from a CAD/FEA-style field export with genuine spatial structure to the full inhomogeneous-shift budget of a real atomic cloud (the mean shift together with the spread across the ensemble, the resulting dephasing time T_2^*, and the clock transition's line profile), computed by the same pipeline validated in Sec. IV, with the trajectory and rotor evaluation modes checked directly against each other on the identical ensemble. This is a demonstration scenario, built the same way Sec. IV C's NPL case and this paper's closed-form validation suite are: every input is a chosen, documented scenario parameter, not sourced from an independent field measurement (Sec. VI states what that stronger case would still require). The scale is anchored to a documented class of real events: Lodewyck et al. observed and then cancelled a stray-charge DC-Stark shift at an operating Sr lattice clock [25], and a field event of this scenario's magnitude would dominate the clock's coherence outright, with nothing in the unmodeled physics competing at that scale. The scenario therefore answers the two questions a practitioner actually brings to it: what signature a charge-patch event would leave on an operating clock's shift, spread, and T_2^*, and what stray-field level a chamber geometry under design can tolerate before it matters.

A. Scenario: an asymmetric two-electrode chamber with a patch spot

The field is a finite-difference solution of Laplace's equation (`examples/generate_showcase_field.py`; the same from-scratch, NumPy/SciPy-only conjugate-gradient solver as this project's earlier COMSOL-format round-trip test, with no COMSOL or other paid software involved) for a grounded chamber containing two electrode plates of *different* size and voltage at different, non-mirror-symmetric positions, plus a small circular patch-potential spot embedded in one wall (a crude but standard model of trapped-charge contamination on a chamber surface). Unlike a simple parallel-plate capacitor (whose field is nearly linear across any small interior region), this scenario is chosen specifically to carry real curvature and cross-terms: the exported field region shows a 7.1% relative magnitude variation over one cloud standard deviation (0.82 mm, two field-grid cells at this scenario's 0.41 mm spacing) from the trap center along its steepest axis. This is a genuinely varying field. Figure 6(a) shows the whole-chamber field magnitude with both electrodes and the patch spot marked; the field at the trap center itself is $|E| = 117.11$ V/m.

Scale reasoning (why the cloud is chamber-scale, not lattice-site scale). A real optical-lattice-site cloud's motional ground-state extent is tens of nanometers (the harmonic-oscillator scale $\sqrt{\hbar/(2m\omega)}$, at a typical lattice-site trap frequency $\omega \sim 10^5$ rad/s), far smaller than any chamber-scale field feature, so it would never sample genuine field curvature; a field-gradient map's real spatial content only shows up in an ensemble whose own spatial extent is a meaningful fraction of the field's variation scale. This showcase therefore uses a classical, chamber-scale trapping stage (e.g. a large-volume "science" dipole or magnetic trap) with a 100 rad/s isotropic trap and a 50 μK, genuinely laser-cooled-scale ^{87}Sr ensemble (100 classical Monte Carlo atoms), giving a classical thermal cloud size $\sigma = \sqrt{k_B T/(m\omega^2)} = 692\,\mu$m, a sizable fraction of the field's own spatial-variation scale, so the ensemble's trajectories genuinely sample real field curvature. This is a scenario-*geometry* choice (trap frequency and temperature), documented exactly as such; no physical constant is touched to reach it.

B. Trajectory and rotor modes, run head to head

Each atom's Sr-87 classical trajectory (Maxwell–Boltzmann velocities, velocity-Verlet propagation in the trap, over a genuine 0.200 s window spanning several trap oscillation periods) is propagated through the fitted field and accumulates its DC-Stark phase via the pipeline's **trajectory mode** (Sec. II B), exactly as `examples/showcase_gradient_dispersion_sr87.yaml` runs through the standard `cliffordclock` CLI/API. The identical Monte Carlo trajectories are then re-accumulated through the **rotor mode**'s Cl(1,3) engine, driven directly against the same trajectories (the shipped config schema's `integration.mode=worldline` keyword exposes the rotor cross-check for static lattice nodes only, Sec. II B; the identical accumulator function also runs, unchanged, against a moving classical trajectory, which is what this figure exercises). **The two evaluations agree to 1.237×10^{-16} relative in the ensemble mean shift, to 2.910×10^{-11} absolute (float64-noise scale) in every individual atom's accumulated phase, and to 1.726×10^{-31} absolute in every individual atom's fractional shift**. That is the message this section exists to make concrete: the trajectory mode is not an approximation of a "more correct" rotor calculation, and the rotor mode is not required to get a spatially varying field's shift budget right; both are already the same number, verified on this exact scenario as well as on the simpler cases of Table III.

C. The inhomogeneous shift budget

The pipeline reports a mean fractional shift $\langle \Delta\nu/\nu_0 \rangle = -9.965 \times 10^{-17}$ with a population spread of 6.514×10^{-18} across the ensemble (SEM 6.514×10^{-19}). The spread is the physically meaningful dispersion quantity here; the SEM in parentheses is only the statistical precision of the mean. The ensemble's accumulated phase spread σ_Φ implies an inhomogeneous dephasing time via the standard Gaussian-dephasing form

$$T_2^* = \frac{\sqrt{2}\,T}{\sigma_\Phi}, \tag{12}$$

(E27, CONVENTIONS.md), giving $T_2^* = 279.7\,\mu$s, roughly three orders of magnitude below this scenario's own interrogation window. An experiment run this way would lose coherence to field-gradient inhomogeneity long before the interrogation time itself, a real, quantitative finding a lab could not get from a field-gradient map and a mean-shift calculation alone. (This number is set by where each atom sits in the field: the mean phase and its spread both scale with the interrogation window here, so T_2^* reflects the ensemble's spatial distribution through the field regardless of the particular window chosen; see `examples/showcase_gradient_dispersion_sr87.yaml`'s own scale-reasoning note.) Figure 6(c) shows the resulting per-atom shift distribution and spectral line profile, with T_2^* annotated; the rotor-mode line profile is overlaid and indistinguishable from the trajectory-mode one at this resolution (rotor $T_2^* = 279.7\,\mu$s).

What a lab could not otherwise easily get. A general-purpose FEA solver reports a field map; converting that map into a gradient-driven dispersion budget for a specific trapped ensemble (spread, T_2^*, and line shape, beyond a mean shift at one point) is exactly the workflow gap Sec. I describes, and exactly what this section demonstrates end to end, with the scalar and rotor engines agreeing on the answer.

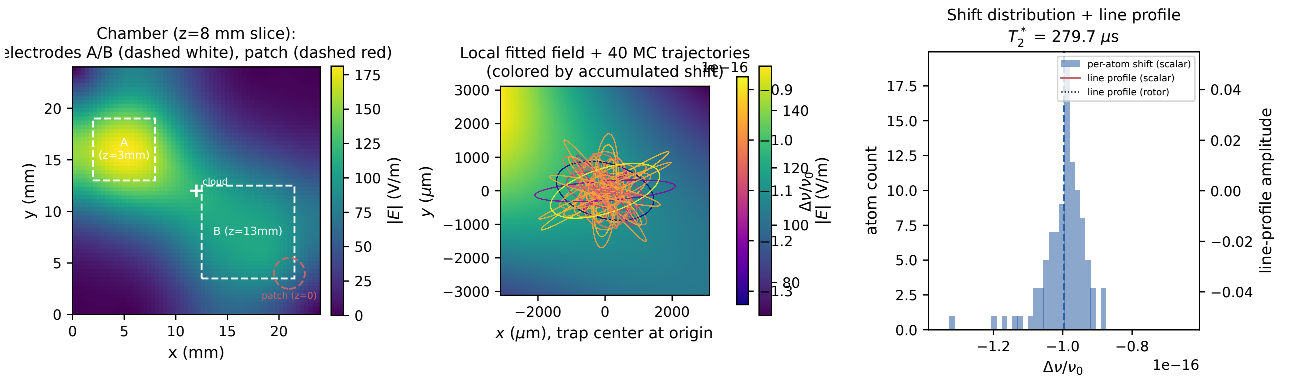

FIG. 6. The showcase scenario (Sec. V). (a) Whole-chamber field magnitude at the trap's z-plane, with the two asymmetric electrodes and the patch-potential wall spot marked (dashed outlines, projected onto this plane; see the electrode/patch z-positions annotated on the panel). (b) The locally fitted field (the same `FieldSmoother` fit the pipeline evaluates) with a subsample of Monte Carlo trajectories overlaid, each colored by its final accumulated fractional shift. (c) The per-atom shift distribution (histogram) and the resulting spectral line profile (both trajectory- and rotor-mode evaluations overlaid), with T_2^* annotated. Regenerated by `paper/figures/fig4_showcase_gradient_dispersion.py` against the real pipeline, using `examples/showcase_gradient_dispersion_sr87.yaml` and its generating field `examples/showcase_field.txt` (`examples/generate_showcase_field.py`).

VI. LIMITATIONS AND ROADMAP

The current physics package covers the field-driven and relativistic terms of an optical-clock frequency budget end to end: DC-Stark shifts from static and gradient fields, electric-quadrupole shifts for ion clocks, blackbody radiation at a uniform temperature, second-order Doppler, and the gravitational redshift across an extended atomic sample. Every one of these terms runs through the same numerics Sec. III pins, so a lab can add a new field geometry or species and get a full dispersion budget from the same validated machinery. Table I states the remaining boundary directly, and every item outside it is a documented, intentional choice with a stated reason. An independent benchmarking pass against two published clock uncertainty budgets [2, 3] found most published systematic line items outside this project's scope, with blackbody radiation dominating both budgets by three or more orders of magnitude; that finding is what ordered the scope extensions shipped so far, and it orders the queue below.

Lattice light shifts are the next physics package on that queue, because they complete the lattice-clock systematic triad alongside BBR and the field-driven terms already covered. Modeling them at the magic wavelength needs species-specific AC polarizability data: the hyperpolarizability, the E2/M1 polarizabilities, and the motional-state intensity-sampling dependence that recent dual-ensemble evaluations resolve directly. Tensor polarizability and RF/micromotion dynamics for ion traps arrive together as one package, because the published Sr^+ measurements of Sec. II F [16] show micromotion-driven tensor Stark shifts dominating the m_J-dependent budget; shipping tensor polarizability without the time-dependent RF field that drives micromotion would model the smaller term while implying full coverage of the larger one. That same time-dependent field machinery, built once, also sets up probe-beam AC Stark support broadly.

Two further packages wait on partner-lab demand. Density and collisional shifts are a different physics class altogether: they need an interaction model between atoms, with s-wave and p-wave collision physics and excitation-fraction dependence, where this project's ensembles are independent particles by construction; some labs manage this systematic entirely through density control and extrapolation, in which case it may stay outside this tool's scope indefinitely. Blackbody radiation's remaining spatial refinement is a per-atom solid-angle map across an extended cloud (Sec. II D's own scope boundary), letting each atom's own position within the ensemble see a different mix of a lab's surfaces. E37 already turns a lab's own per-surface solid-angle weights, geometric, FEA-derived, or ray-traced, into an exact per-moment shift and uncertainty band (Sec. II D); a partner lab's own position-resolved exchange-factor output, the same extension Sec. IV J demonstrates on a real published chamber, is what turns that single-environment description into the per-atom map.

The multi-surface thermal environment and the quantum-motional systematics package (Secs. II D and II G) bring two systematics into the modeled column of Table I: a field-deployed clock's non-uniform surroundings, and a trapped ion's own motional-state contribution to second-order Doppler, including the two-ion intrinsic-micromotion correction Sec. IV I validates. A per-atom solid-angle map across an extended thermal environment remains open (Sec. II D's own scope boundary): every atom in an ensemble still sees the identical environment description, unlike the extended-lattice regime's own per-site treatment of Eq. (6)'s gravitational term. Crystals with more than two ions need a numeric normal-mode eigensolver, with no closed form in general. The excess-micromotion RF dynamics that produce a trapped ion's stray-field-driven displacement remain future work; Sec. II G's EMM channel takes their measured consequence as a lab-supplied input today. Non-Gaussian motional-state Ramsey visibility, Fock states with $n \geq 1$ and cat states, sits outside the worldline-ensemble representation Sec. II H builds on.

The rotor engine implements the physical DC-Stark coupling directly (the Ω-bivector construction `build_omega_stark`, Sec. II B), with its own dedicated, direct verification against the scalar path (including a re-run of the NPL reproducibility case through the rotor integrator and the showcase's head-to-head Monte Carlo comparison, Sec. V). By design, the fast analytic and trajectory modes' performance tiers (`fast_path`, `direct`, `secular`; Table II) stay on the scalar accumulator: they are built around a coupling-agnostic rate function, and routing them through a per-step Cl(1,3) rotor construction would trade away their performance advantage for no accuracy benefit, given the scalar/rotor equivalence Sec. II B establishes for every physical coupling this project currently implements. That equivalence also locates the rotor engine's roadmap value precisely: it begins where a future extension's physics stops being expressible as the single scalar pivot this section's other modes assume. Magnetic couplings (Zeeman shifts, Table I), internal-state structure beyond a two-level clock transition, and transport-induced geometric phases (a rotor accumulated along the moving trajectory itself) are the concrete candidates, and each needs the rotor's bivector/spinor structure to be represented at all.

The clearest concrete next step toward a genuine **blind prediction** case (Sec. IV) is access to a published source that independently characterizes a stray field this project's own pipeline has not already been used, by that same source, to convert to a shift. One such candidate is known by citation (Li et al. 2024 [26], cited by Jia et al. 2026 [3] as the source that originally characterized an applied field underlying one of their systematic-shift budget entries) but was not accessible to this project through any legitimate open-access route at the time of writing; its content remains unknown and is neither ruled in nor ruled out as a future reproducibility or blind-prediction case. A structured collaboration with a clock-apparatus group willing to supply a field characterization ahead of (or independent of) their own shift measurement, a beta blind-prediction program, is the concrete path to closing this gap, and is future work; no such collaboration is named here pending the owner's own outreach.

VII. CODE AVAILABILITY AND CITATION

The source code, test suite, documentation, and the figure-generation scripts underlying this paper are available at `https://github.com/velar-mbr/CliffordClock` under the GNU Affero General Public License v3.0 or later (AGPLv3). The Python package and command-line interface are both distributed under the name `cliffordclock` (pip package and CLI command). A machine-readable citation file (`CITATION.cff`) is provided in the repository root.

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